# Section 6: Retrieval Augmented Generation (RAG)

# 6.1. Understanding RAG

# LLM Challenges

## Traditional LLMs

Large Language Models (LLMs) face key limitations when relying only on pre-trained data without external knowledge access during inference.

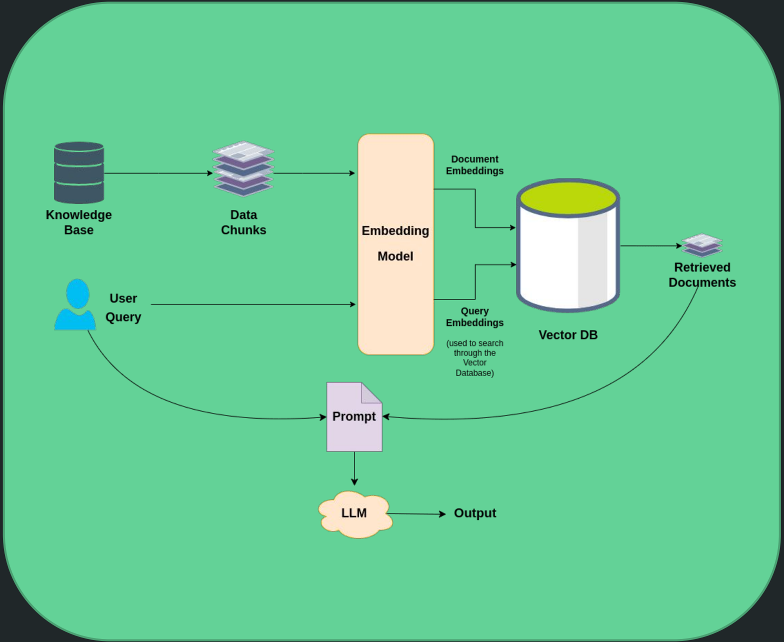
- May produce incorrect or fabricated information when answers are missing
- Often provide outdated, generic, or low-specificity responses
- Generate outputs based on non-verifiable or non-authoritative sources

## RAG Solutions

Retrieval Augmented Generation (RAG) overcomes these challenges by connecting LLMs to trusted external knowledge sources in real-time.

- Ensures access to the most current and accurate information
- Supports source attribution and fact verification
- Eliminates costly retraining while maintaining scalability and reliability

# About RAG




## Retrieval Augmented Generation

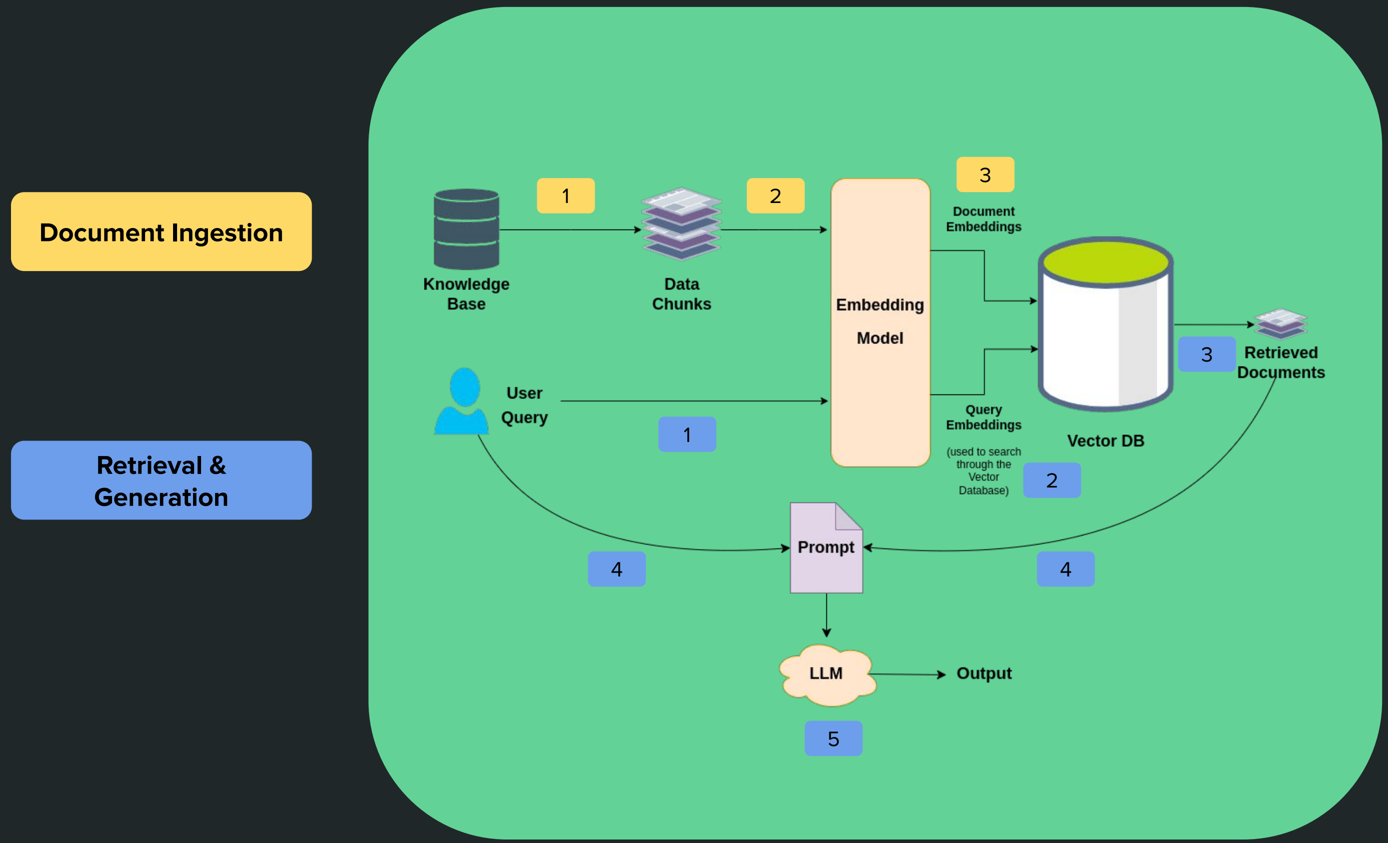
### Core Concept

RAG enhances large language model outputs by retrieving and grounding them in authoritative external knowledge sources, improving accuracy, reliability, and factual consistency.

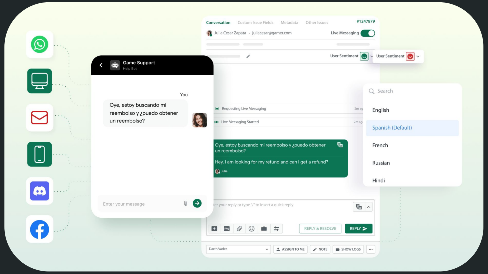
### Key Benefits

RAG brings in up-to-date, domain-specific knowledge for relevant answers.It reduces hallucinations and improves reliability.
It also ensures source traceability across domains.

# RAG Architecture



# RAG Applications

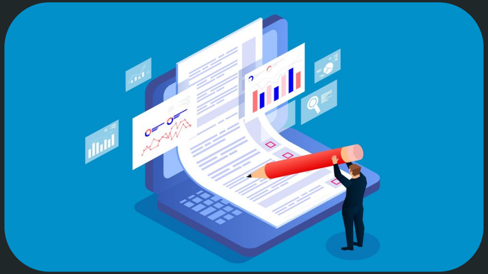



## Customer Support

Automated chatbots provide accurate company-specific answers from knowledge bases and documentation

## Knowledge Management

Internal Q&A systems enable employees to access HR policies compliance documents efficiently



## Search Enhancement

LLM-augmented search engines provide contextual answers alongside traditional search results

# Vector DataBases

| Database | Type | Hybrid | Perf | Best For |
|---|---|---|---|---|
| **Pinecone** | Cloud | Limited | 🔥🔥🔥 | Scale, no ops |
| **Weaviate** | OSS+Cloud | Yes | 🔥🔥🔥 | Hybrid search |
| **Milvus** | OSS+Cloud | Partial | 🔥🔥🔥🔥 | Billion data |
| **Qdrant** | OSS+Cloud | Partial | 🔥🔥🔥 | Fast + filters |
| **pgvector** | OSS (SQL) | SQL | 🔥🔥 | In-DB vectors |
| **Elastic/Open** | OSS+Cloud | Yes | 🔥🔥🔥 | Enterprise use |
| **Redis** | OSS+Cloud | Basic | 🔥🔥 | In-memory |
| **Vespa** | OSS+Cloud | Yes | 🔥🔥🔥 | Complex rank |
| **ChromaDB** | OSS | Basic | 🔥 | Prototype |